Reg. No. | | | | | | | | | | | [14EE65]

SIXTH SEMESTER B.Tech EXAMINATION, JUNE–2019

POWER PLANT ENGINEERING

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by selecting atleast **TWO** questions from each part.
2. Missing data may be suitably assumed.

PART-A

1.	a.	With a neat block diagram explain the working of solar power plant.	06
	b.	Explain the concept of cogeneration.	04
	c.	Mention the merits and demerits of wind power plant.	04
2.	a.	With a neat schematic diagram, explain the essential elements of high head hydroelectric power plant.	09
	b.	Enumerate on the factors to be considered for the selection of site hydroelectric power plant.	05
3.	a.	With a neat diagram briefly explain water tube boiler.	07
	b.	Explain with a line diagram, fuel handling of thermal power plant.	07
4.	a.	Discuss some of the safety measures incorporated for nuclear power plant.	04
	b.	Explain with a neat sketch pressurized water reactor. Mention their advantages and disadvantages.	10

PART-B

5.	a.	Describe briefly the main components of diesel electric plant with a block diagram.	10
	b.	Compare open cycle and closed cycle gas turbine plants.	04
6.	a.	What is power factor? Explain any one method of improving power factor.	06
	b.	An electric supply company having a maximum load of 50 MW generates 18 X 10^7 units per annum and supply consumers have an aggregate demand of 75 MW. The annual expenses including capital charges are: For fuel = Rs 90 lakhs Fixed charges concerning generation = Rs 28 lakhs Fixed charges concerning transmission and distribution = Rs 32 lakhs Assuming 90% of the fuel cost is essential to running charges and the loss in transmission and distribution as 15% of kWh generated, deduce a two part tariff to find the actual cost of supply to the consumers.	08

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7. a. Give the comparison of outdoor and indoor substations. **04**

b. What is a transformer substation? What are the different types of transformer substation? Illustrate the answer with a suitable block diagram. **10**

8. a. Describe in detail the working of grounding transformer with necessary circuit diagrams. **08**

b. A 132 kV, 3 phase, 50 Hz transmission line 200 km long consists of 3 conductors of effective diameter 20 mm arranged in a vertical plane with 4 m spacing and regularly transposed. Find the inductance and KVA rating of the arc suspension coil in the system. **06**